

MACHINE LEARNING

UNIT-1

Introduction: Introduction to Machine learning, Supervised learning, Unsupervised learning, Reinforcement learning. Deep learning.

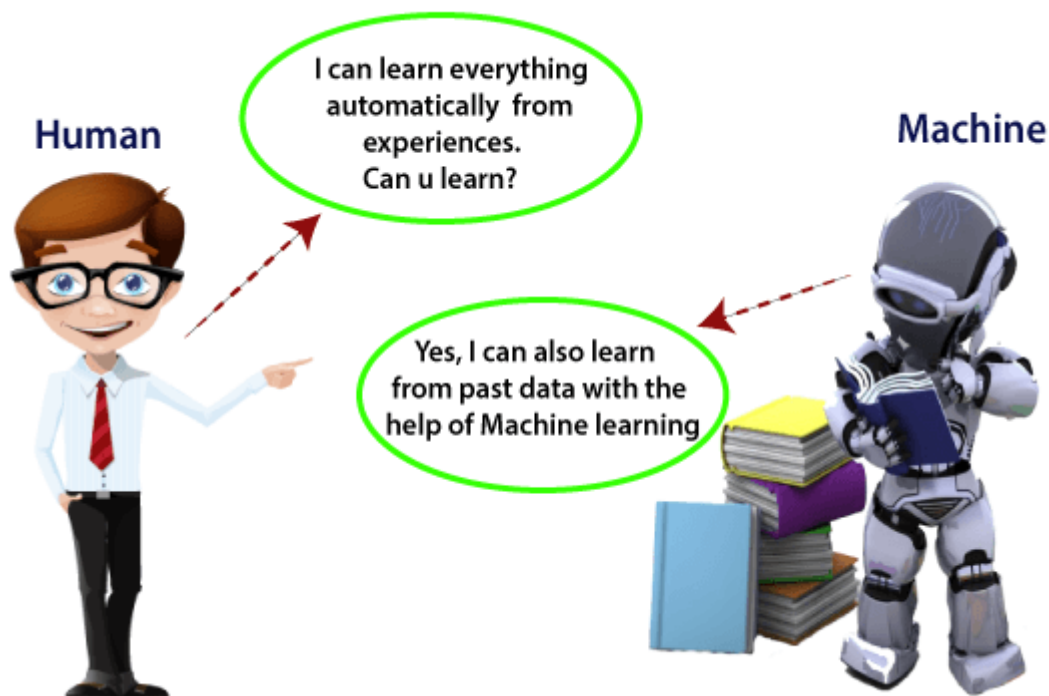
Feature Selection: Filter, Wrapper, Embedded methods.

Feature Normalization:- min-max normalization, z-score normalization, and constant factor normalization

Introduction to Dimensionality Reduction: Principal Component Analysis (PCA), Linear Discriminant Analysis (LDA)

What is Machine Learning?

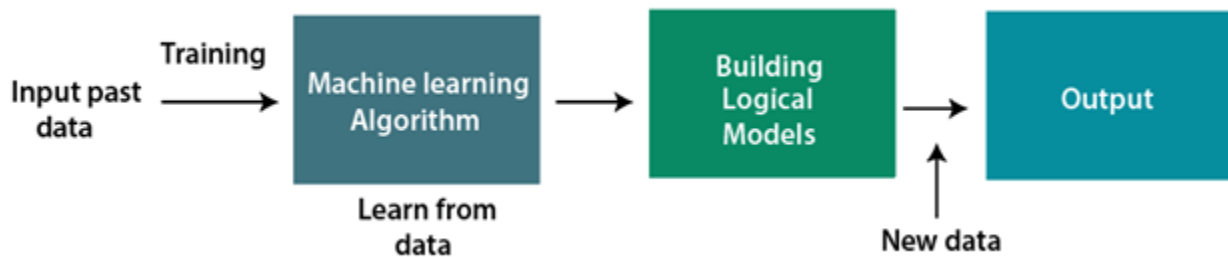
- Humans learn everything from their experiences with their learning capability Computers work on our instructions.
- But can a machine also learn from experiences or past data like a human does? So here comes the role of **Machine Learning**.



- Machine Learning is said as a subset of **artificial intelligence** that is mainly concerned with the development of algorithms which allow a computer to learn from the data and past experiences on their own.

How does Machine Learning work

- A **Machine Learning system** learns from historical data, builds the prediction models, and whenever it receives new data, predicts the output for it.
- The accuracy of predicted output depends upon the amount of data, as the huge amount of data helps to build a better model which predicts the output more accurately.
- Suppose we have a complex problem, where we need to perform some predictions, so instead of writing a code for it, we just need to feed the data to algorithms, and with the help of these algorithms, machine builds the logic as per the data and predict the output. Machine learning has changed our way of thinking about the problem.



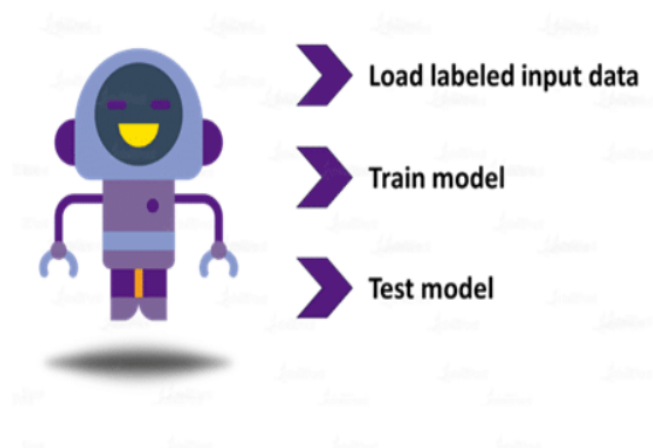
Examples of machine learning :

- Smartphones detecting faces while taking photos or unlocking themselves.
- Facebook, LinkedIn or any other social media site recommending your friends and ads you might be interested in.
- Amazon recommending you the products based on your browsing history.
- Banks using Machine Learning to detect Fraud transactions in real-time.



Supervised Learning: What is it?

Consider yourself as a student sitting in a math class wherein your **teacher is supervising** you on how you're solving a problem or whether you're doing it correctly or not. This situation is similar to what a **supervised learning algorithm** follows, i.e., with input provided as a **labeled dataset**, a model can learn from it. Labeled dataset means, for each dataset given, an **answer or solution** to it is given as well. This would help the model in learning and hence provide the result of the problem easily.



So, a labeled dataset of animal images would tell the model whether an image is of a dog, a cat, etc. Using which, a model gets training, and so, whenever a new image comes up to the model, it can compare that image with the labeled dataset for predicting the correct label.

Supervised Learning Problems

There are two types of problems:

1. **classification problems**
2. **regression problems.**

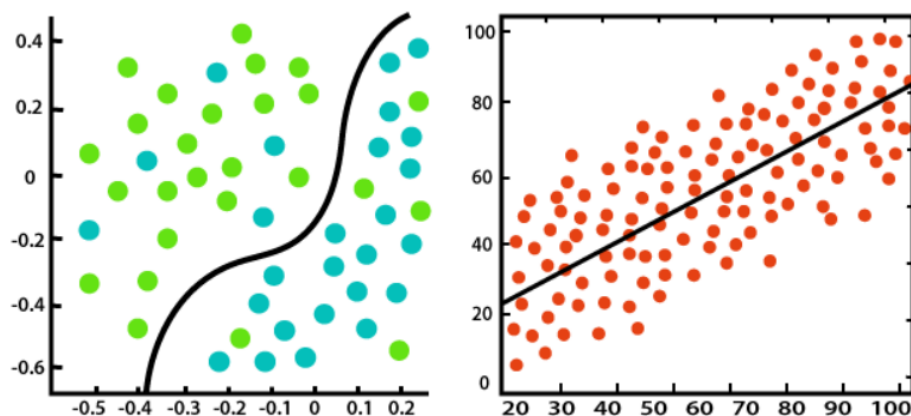
Classification problems ask the algorithm to predict a discrete value that can identify the input data as a member of a particular class or group. Taking up the animal photos dataset, each photo has been labeled as a dog, a cat, etc., and then the algorithm has to **classify** the new images into any of these **labeled categories**.

Regression problems are responsible for continuous data, e.g., for predicting the price of a piece of land in a city, given the area, location, etc.. Here, the input is sent to the machine for predicting the price according to previous instances. And the machine determines a function that would map the pairs.

Regression and Classification algorithms are Supervised Learning algorithms. Both the algorithms are used for prediction in Machine learning and work with the labeled datasets. But the difference between both is how they are used for different machine learning problems.

The main difference between Regression and Classification algorithms that Regression algorithms are used to **predict the continuous** values such as price, salary, age, etc. and Classification algorithms are used to **predict/Classify the discrete values** such as Male or Female, True or False, Spam or Not Spam, etc.

Consider the below diagram:



Classification:

Classification is a process of finding a function which helps in dividing the dataset into classes based on different parameters. In Classification, a computer program is trained on the training dataset and based on that training, it categorizes the data into different classes.

The task of the classification algorithm is to find the mapping function to map the input(x) to the discrete output(y).

Example: The best example to understand the Classification problem is Email Spam Detection. The model is trained on the basis of millions of emails on different parameters, and whenever it receives a new email, it identifies whether the email is spam or not. If the email is spam, then it is moved to the Spam folder.

Types of ML Classification Algorithms:

Classification Algorithms can be further divided into the following types:

- Logistic Regression
- K-Nearest Neighbours
- Support Vector Machines
- Naïve Bayes
- Decision Tree Classification

Regression:

Regression is a process of finding the correlations between dependent and independent variables. It helps in predicting the continuous variables such as prediction of **Market Trends**, prediction of House prices, etc.

The task of the Regression algorithm is to find the mapping function to map the input variable(x) to the continuous output variable(y).

Example: Suppose we want to do weather forecasting, so for this, we will use the Regression algorithm. In weather prediction, the model is trained on the past data, and once the training is completed, it can easily predict the weather for future days.

Types of regression algorithms

- Simple Linear Regression
- Multiple Linear Regression

Regression Algorithm	Classification Algorithm
In Regression, the output variable must be of continuous nature or real value.	In Classification, the output variable must be a discrete value.
The task of the regression algorithm is to map the input value (x) with the continuous output variable(y).	The task of the classification algorithm is to map the input value(x) with the discrete output variable(y).
Regression Algorithms are used with continuous data.	Classification Algorithms are used with discrete data.
In Regression, we try to find the best fit line, which can predict the output more accurately.	In Classification, we try to find the decision boundary, which can divide the dataset into different classes.
Regression algorithms can be used to solve the regression problems such as Weather Prediction, House price prediction, etc.	Classification Algorithms can be used to solve classification problems such as Identification of spam emails, Speech Recognition, Identification of cancer cells, etc.

Unsupervised Learning

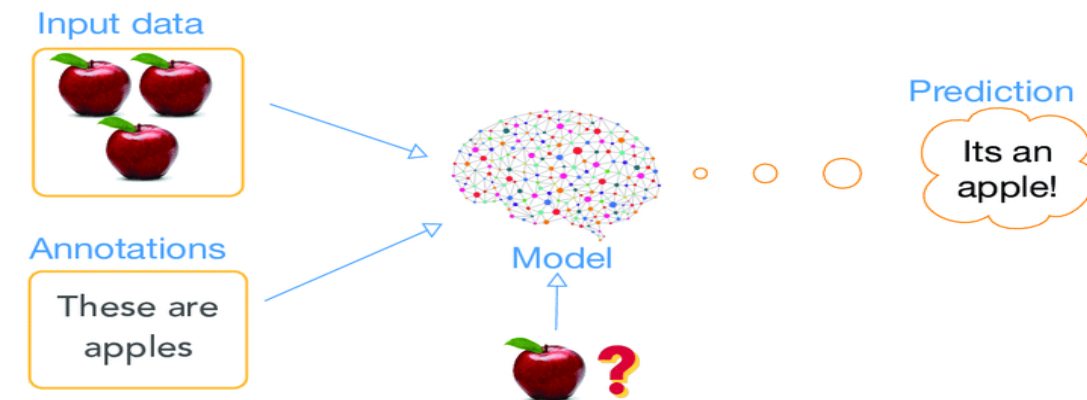
- In supervised learning, the dataset is properly labeled, meaning, a set of data is provided to train the algorithm. The major difference between supervised and unsupervised learning is that there is **no complete and clean labeled dataset in unsupervised learning**.
- Unsupervised learning is a type of **self-organized learning** that helps find previously unknown patterns in data sets **without pre-existing labels**.
- Consider the animal photo example used in supervised learning. Suppose, there is **no labeled dataset** provided. Then, how can the model find out if an animal is a cat or a dog or a bird? Well, if the model has been provided some information such as if an animal has feathers, a beak, wings, etc. it is a bird. In the same way, if an animal has fluffy fur, floppy ears, a curly tail, and maybe some spots, it is a dog, and so on.
- The goal of unsupervised learning is to restructure the input data into new features or a group of objects with similar patterns.

Unsupervised learning is classified into two categories of algorithms:

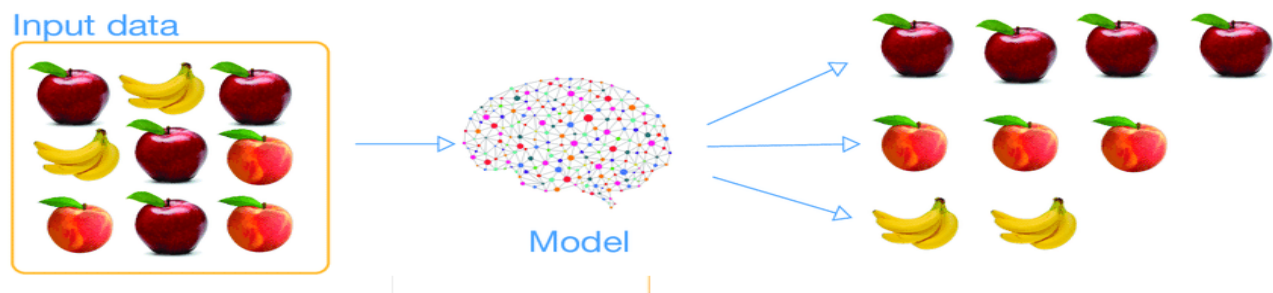
- **Clustering:** A clustering problem is where you want to discover the inherent groupings in the data, such as grouping customers by purchasing behavior.
- **Association:** An association rule learning problem is where you want to discover rules that describe large portions of your data, such as people that buy X also tend to buy Y.



supervised learning



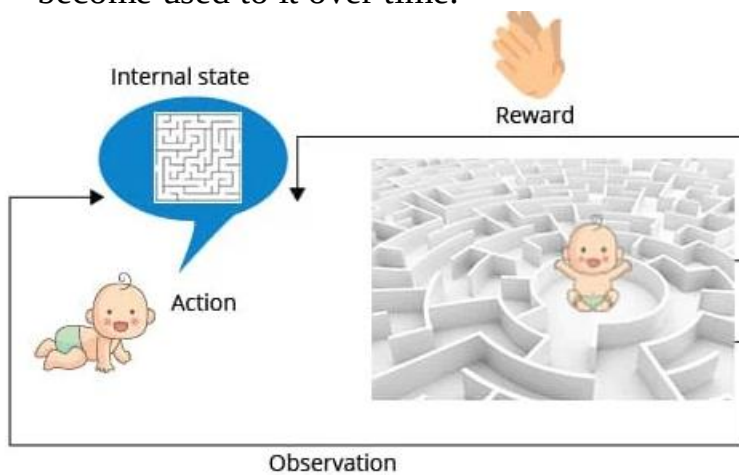
unsupervised learning



Reinforcement Learning

child trying to take his/her first steps. What will be the instructions he/she follows to start walking?

1. **Observing** others walking and trying to replicate the same
2. **Standing still**
3. **Remaining still**
4. Trying to **balance** the body weight, along with deciding on which foot to advance first to start walking. It sounds like a difficult and challenging task for a child to get up and walk, right? But for us, it is easy since we have become used to it over time.

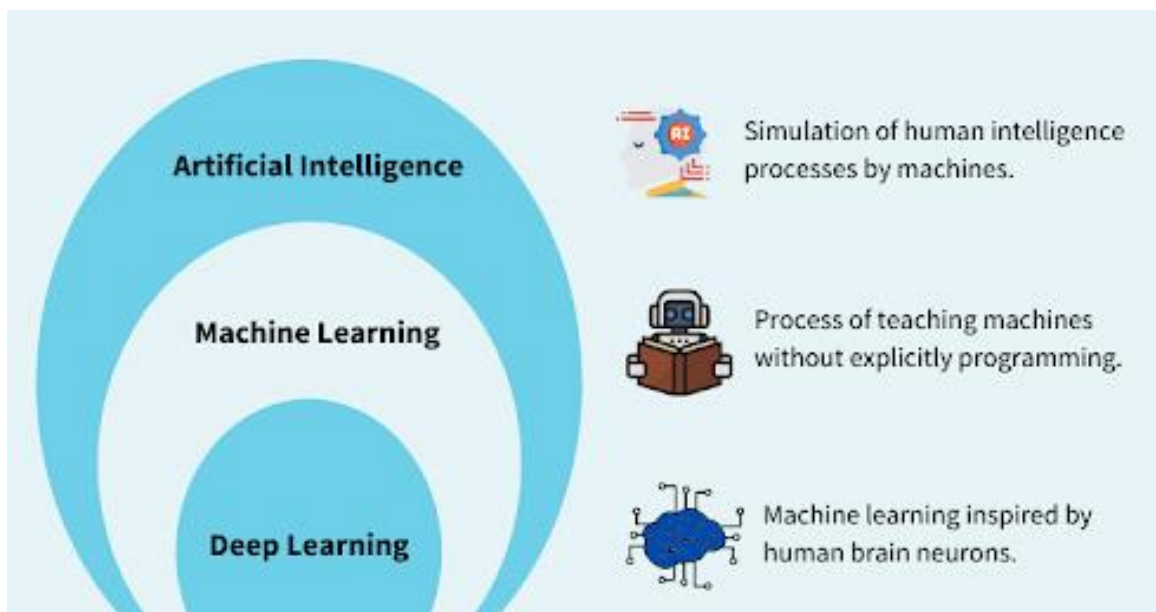


Now, putting it together, **a child is an agent** who is trying to manipulate the **environment** (surface or floor) by trying **to walk** and going from **one state to another** (taking a step). A child **gets a reward** when he/she takes a **few steps** (appreciation) but will **not receive any reward or appreciation** if he/she is unable to walk. This is description of a **reinforcement learning problem**.

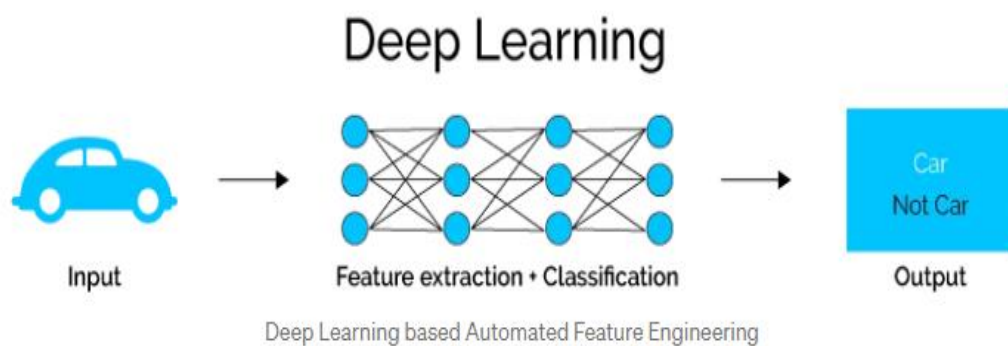
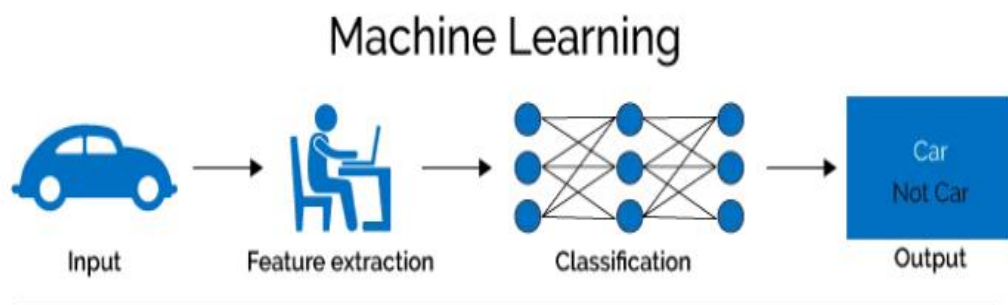
Criteria	Supervised Learning	Unsupervised Learning	Reinforcement Learning
Definition	The machine learns by using labeled data	The machine is trained on unlabeled data without any guidance	An agent interacts with its environment by performing actions & learning from errors or rewards
Type of problems	Regression & classification	Association & clustering	Reward-based
Type of data	Labeled data	Unlabeled data	No predefined data
Training	External supervision	No supervision	No supervision
Approach	Maps the labeled inputs to the known outputs	Understands patterns & discovers the output	Follows the trial-and-error method

- **Supervised learning** is when a model learns from a labeled dataset with guidance. And,
- **Unsupervised learning** is where the machine is given training based on unlabeled data without any guidance.
- **Reinforcement learning** is when a machine or an agent interacts with its environment, performs actions, and learns by a trial-and-error method.

Deep learning



- AI is the superset, ML comes under the AI and deep learning comes under the ML.
- **Artificial Intelligence**, it is to automate human tasks and to develop intelligent machines that could learn without human intervention.
- It deals with making the machines smart enough so that they can perform those tasks which normally require human intelligence.
- Self-driving cars are the best example of artificial intelligence. These are the robot cars that can sense the environment and can drive safely with little or no human involvement.
- **Machine learning** is the subfield of Artificial Intelligence. Have you ever thought about how youtube knows which videos should be recommended to you? How does Netflix know which shows you'll most probably love to watch without even knowing your preferences? The answer is machine learning.



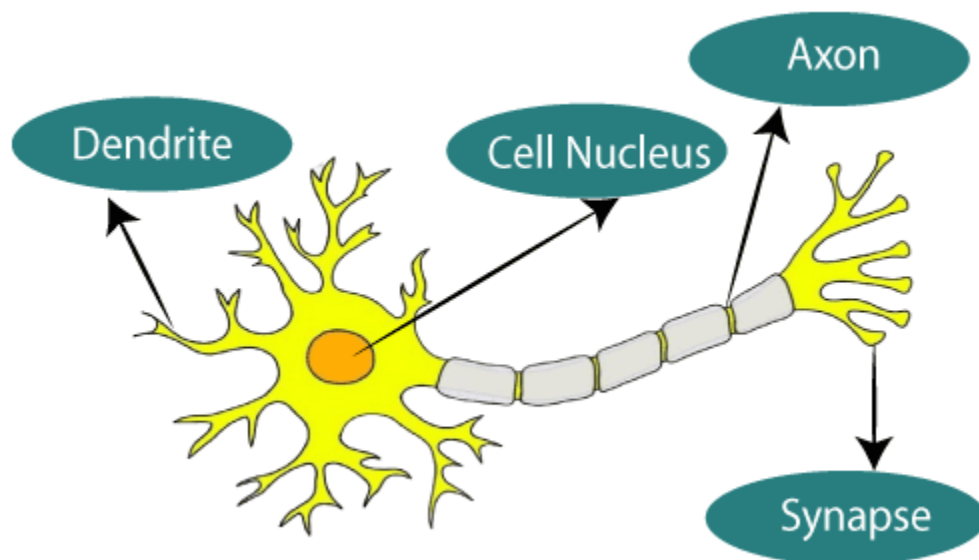
	Machine Learning	Deep Learning
Definition	Subfield of AI that focuses on machines being able to learn without being explicitly programmed.	Subfield of ML that focuses on machines being able to mimic the human brain to perform highly complex AI problems.
Data Feeding	We give structured data to the machine that builds the ML model.	We give unstructured data or you can say the raw input to the neural network.
Volume of Data	ML models deal with datasets having thousands of data rows.	Deep learning models mostly deal with datasets having millions of data rows.
Training Time	ML models take less time in training because of the small data size.	It takes a huge amount of time because of massive data points.

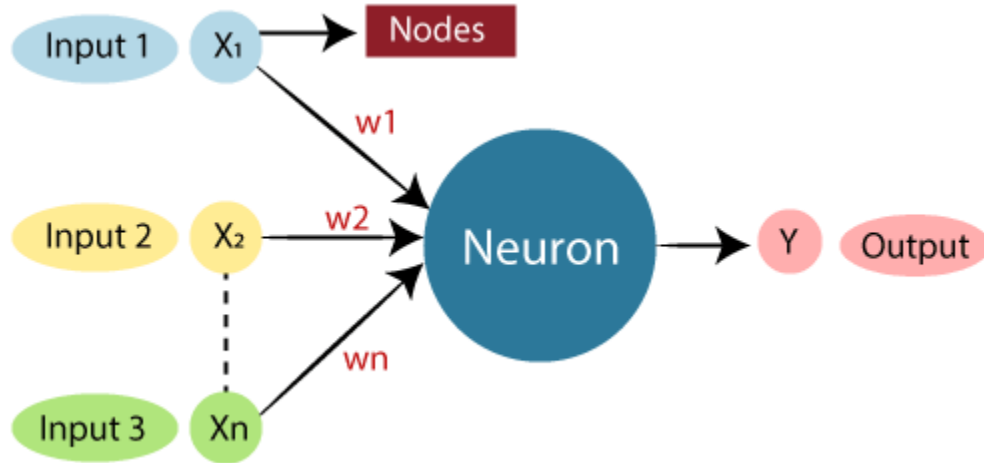
Human Involvement	Machine learning models are easy to build but require more human interaction to make better predictions.	Deep learning models are difficult to build as they use complex multilayered neural networks but they have the capability to learn by themselves.
Feature Engineering	Feature engineering is done explicitly by humans.	No need of feature engineering, neural networks automatically detect important features.
Goal	To give the output as close as it can be to the expected output.	To mimic the human brain processing, how they actually think. If somehow machines are able to think that way they will automatically generate the right output.
Interpreting Results	It is easy to explain the results of an ML model.	It is difficult to explain the results of a deep learning model since it's hard to interpret the output of a complex multi-layered neural network.

Performance	ML models show good performance on small and medium-sized datasets.	Deep learning models show better performance on huge datasets.
Applications	Fraud detection, Recommendation systems, Pattern recognition, and so on.	Customer support, Image processing, Speech recognition, Object recognition, Natural language processing, computer vision, and so on.

Artificial Neural Network

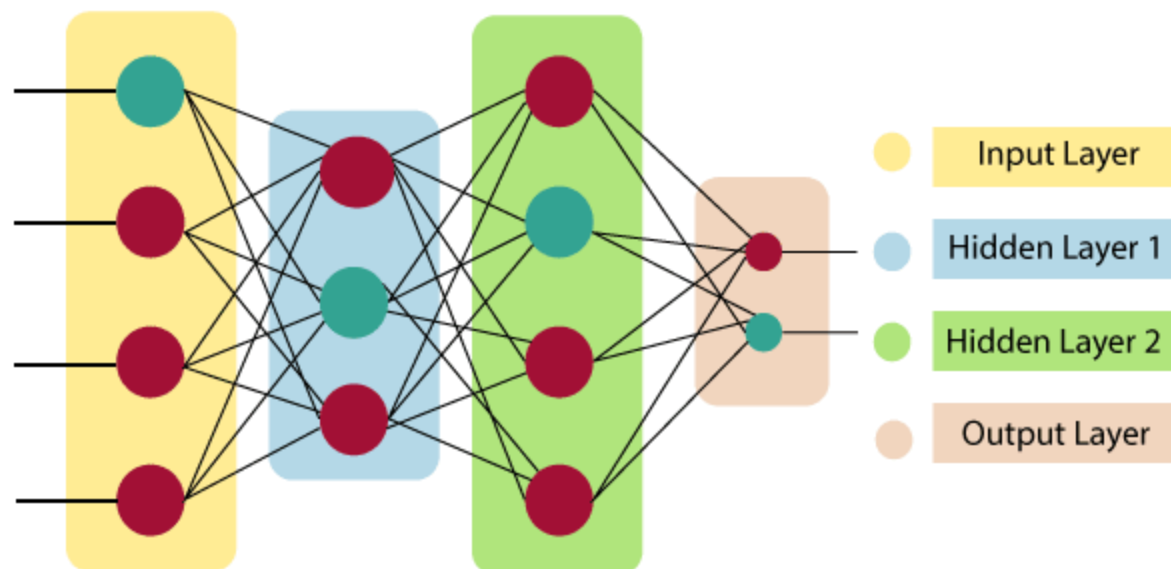
The term "**Artificial Neural Network**" is derived from Biological neural networks that develop the structure of a human brain. Similar to the human brain that has neurons interconnected to one another, artificial neural networks also have neurons that are interconnected to one another in various layers of the networks. These neurons are known as nodes.





Relationship between Biological neural network and artificial neural network:

Biological Neural Network	Artificial Neural Network
Dendrites	Inputs
Cell nucleus	Nodes
Synapse	Weights
Axon	Output



Input Layer:

As the name suggests, it accepts inputs in several different formats provided by the programmer.

Hidden Layer:

The hidden layer presents in-between input and output layers. It performs all the calculations to find hidden features and patterns.

Output Layer:

The input goes through a series of transformations using the hidden layer, which finally results in output that is conveyed using this layer.

The artificial neural network takes input and computes the weighted sum of the inputs and includes a bias. This computation is represented in the form of a transfer function.

$$\sum_{i=1}^n w_i * x_i + b$$

It determines weighted total is passed as an input to an activation function to produce the output. Activation functions choose whether a node should fire or not. Only those who are fired make it to the output layer.

suppose you want to go to USA but only if:

- x_1 -> The airline ticket is less than \$1,000.
- x_2 -> Your friend can go with you.

You represent this decision with this simple vector of possible inputs:

(1,0), (0,1), (1,1), and (0,0).

In the first case (1,0) the ticket is > 1,000 and your friend cannot go with you.

You put some weight on each of these two calculations. For example, if you are on a budget and cost is important, give it weight $w_1=4$. And whether your friend can go or not is not as important. So give it a weight of $w_2=3$.

So you have this function for **USA**:

$$(x_1 * w_1) + (x_2 * w_2) = (x_1 * 4) + (x_2 * 3) > \text{some threshold, } b, \text{ say, } 4.$$

We move b to the other side and write:

If $(x_1 * 4) + (x_2 * 3) - 4 > 0$ then Go to USA (i.e., perceptron says 1)-

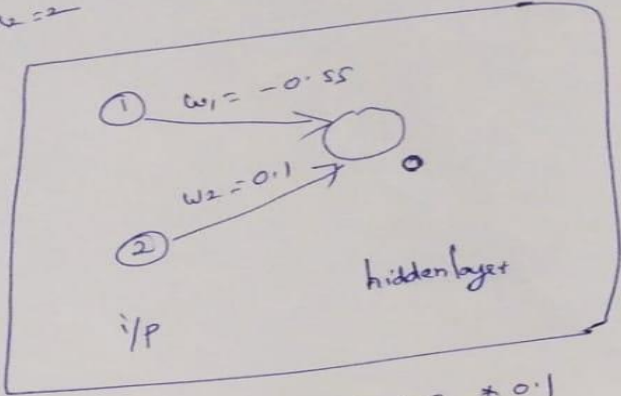
Then feed vectors into the equation. Obviously if the ticket is > \$1,000 and if your friend cannot go (0,0) then you will not make the trip, because $(0 * 3) + (0 * 4) - 4$ is obviously < 0 .

If the ticket is cheap but you are going alone then go anyway:

$$(1 * 4) + (0 * 3) - 4 = 0 \text{ which is not bigger than } 0.$$

Bias & Threshold

$x_1 = 1$
 $x_2 = 2$



$$\sum w_i x_i = 1 * -0.55 + 2 * 0.1$$
$$= -0.35$$

apply Activation function
$$= \max(0, -0.35)$$
$$= 0$$

So, 0, No information passed to next layer

So, take threshold (at certain point we can take decision)

take Threshold = -1 (random value)

bias is opposite to Threshold.

$$= 1 * -0.55 + 2 * 0.1 + 1$$
$$= 0.65$$

Apply Activation function.

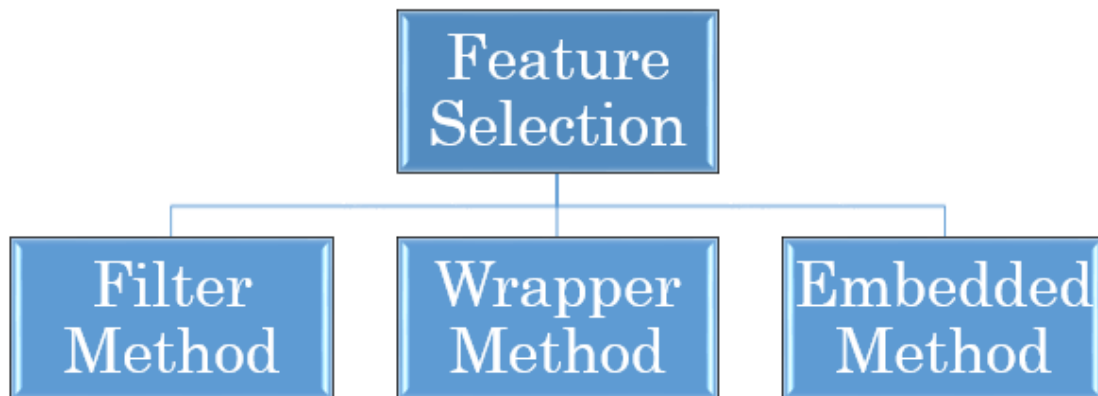
$$= \max(0, 0.65)$$
$$= 0.65 \text{ (Node is fired)}$$

Feature Selection Methods

- identifying the important features from a set of given data and removing the irrelevant or less important features which do not contribute much to our decision making in order to achieve better accuracy for our model.
- In machine learning and statistics, feature selection, also known as **variable selection**, **attribute selection** or **variable subset selection**, is the process of reducing the number of input variables when developing a predictive model.

Feature selection techniques uses:

- It reduces model complexity by dropping some irrelevant features.
- Helps ML algorithm to train a model faster.
- Selecting the best features helps the model to perform well. For example, Suppose we want to create a model that automatically decides which car should be crushed for a spare part, and to do this, we have a dataset. This dataset contains a Model of the car, Year, Owner's name, Miles. So, in this dataset, the name of the owner does not contribute to the model performance as it does not decide if the car should be crushed or not, so we can remove this column and select the rest of the features(column) for the model building.



1. Filter Method:

- In this method, features are **filtered** based on general characteristics (some metric such as correlation) of the dataset such **correlation with the dependent variable**.
- It is **faster** and usually the better approach when the number of features are huge.
- **Avoids over fitting** but sometimes may fail to select best features.

ATTRIBUTES OR FEATURES					TARGET FEATURES
A (%)	B (Roll No)	C (Maths M)	D (C++ M)	E (Name)	T (Target Variable)
75	1	78	60	Neha	Pass
35	2	20	21	Ajay	Fail
90	3	87	89	Rahul	Pass
39	4	25	22	Gauri	Fail



Filter methods examples

1.Information Gain

It is defined as the amount of information provided by the feature for identifying the target value. Information gain of each attribute is calculated considering the target values for feature selection.

2.Variance Threshold

It removes all features whose variance is lower than threshold.

Drop the features which are having less variance than **the threshold**.

Threshold = 0.2

Experience	Age	Designation
2	25	Data Scientist
2	35	Data Analyst
2	39	HR
2	49	Business Analyst
1	20	Data Scientist
2	51	Data Engineer
2	42	Data Scientist
2	53	Data Analyst
2	32	HR
2	21	Business Analyst

Salary
25000
40000
30000
45000
32000
30000
50000
60000
20000
37000

$$Variance = \sum \left(\frac{Xi - Mean}{n - 1} \right)^2$$

0.09 134.20

Activate Windows

Experience feature is removed.(0.09<0.2)

Age feature is selected.

We can apply variance threshold on numerical features.

3.threshold

Let's take an example of logistic regression applied to spam detection, where the two classes are *spam* and *non-spam*. Logistic regression with the sigmoid returns a probability between 0 and 1 for an input data sample to belong to the positive class. A probability of 0.99 means that the email is very likely to be spam, and a probability of 0.003 that it is very likely to be non-spam. If the probability is 0.51, the classifier is less able immediately to determine the nature of the email.

4.

Correlation Coefficients

Correlation helps measure the linear relationship between two or more features and is primarily valid when features are numeric. Its application extends as a feature selection method since the correlation matrix, a.k.a the heatmap, helps visualize the relationship between the features and the target variable. It can also reveal Information on collinear features, which are redundant in the analysis since they don't contribute to any new information; thus, removing such features is recommended. Secondly, we can decide on a threshold value of correlation. If the absolute correlation between the feature and target variable is lower than that threshold, we can discard that feature from the analysis.

If the **correlation** is lesser than the **Threshold** then drop those features.

Threshold = 0.5

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

Experience	Salary
2	25000
2	35000
2	39000
2	49000
1	18000
2	54000
2	43000
2	55000
2	32000
2	22000

Correlation = 0.13

Age	Salary
25	25000
35	35000
39	39000
49	49000
20	18000
51	54000
42	43000
53	55000
32	32000
21	22000

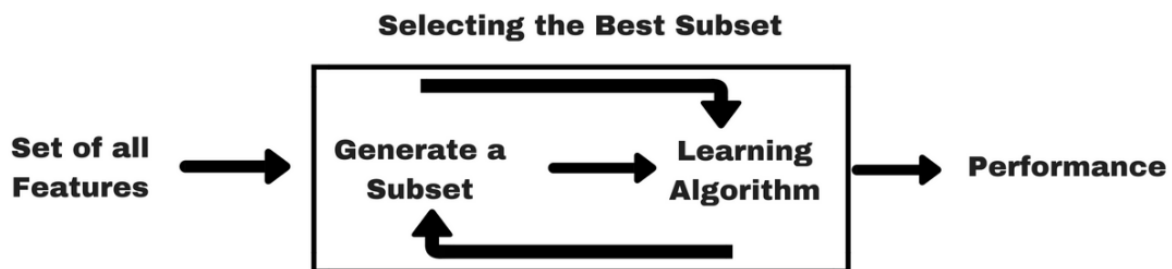
Correlation = 0.99

Activate Windows
Go to Settings to activate Windows.

2. Wrapper Method:

- In wrapper method, the feature selection algorithm exists as a **wrapper** around the **predictive model algorithm** and **uses the same model to select best features**.
- Though computationally **expensive and prone to over fitting**, gives better performance.

Wrapper feature selection methods create many models with different subsets of input features and select those features that result in the best performing model according to a performance metric



$$M1 = A + B$$

$$M2 = A + C + D$$

$$M2 = A + E$$

$$M4 = B + E$$

ATTRIBUTES OR FEATURES					TARGET FEATURES
A (%)	B (Roll No)	C (Maths M)	D (C++ M)	E (Name)	T (Target Variable)
75	1	78	60	Neha	Pass
35	2	20	21	Ajay	Fail
90	3	87	89	Rahul	Pass
39	4	25	22	Gauri	Fail

Wrapper method examples

1. **Forward Selection:** In each iteration, we keep adding the feature which best improves our model till an addition of a new variable does not improve the performance of the model.

We have 4 features

Take f1,buildmodel,calculate accuracy=97

Take f2,buildmodel,calculate accuracy=95

Take f3,buildmodel,calculate accuracy=98

Take f4,buildmodel,calculate accuracy=92

So select f3 feature.

Next iteration

Take f3&f1,buildmodel,calculate accuracy=98

Take f3&f2,buildmodel,calculate accuracy=95

Take f3&f4,buildmodel,calculate accuracy=97

F1,f3 are selected.

This process continues until no improvement of the performance of the model

2. **Backward Elimination:** In backward elimination, we start with all the features and removes the least significant feature at each iteration which improves the performance of the model. We repeat this until no improvement is observed on removal of features.

Take f1,f2,f3,f4,f5 features calculate accuracy=97

Next select f1 & build a model with f2,f3,f4,f5 calculate accuracy=97

Next select f2 & build a model with f1,f3,f4,f5 calculate accuracy=95

Next select f3 & build a model with f2,f1,f4,f5 calculate accuracy=96

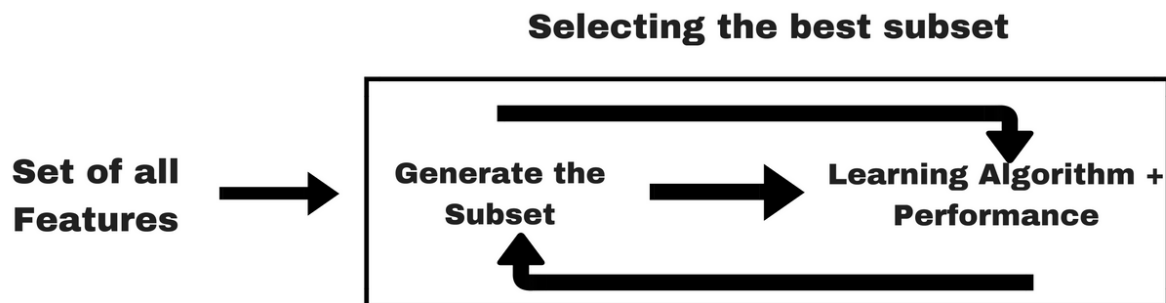
So select only f2,f3,f4,f5

This process continues

3. **Recursive Feature elimination:** It is a greedy optimization algorithm which aims to find the best performing feature subset. It repeatedly creates models and keeps aside the best or the worst performing feature at each iteration. It constructs the next model with the left features until all the features are exhausted. It then ranks the features based on the order of their elimination.

3. Embedded Method:

Embedded methods combine the qualities' of filter and wrapper methods.



LASSO Regularization L1

LASSO Regularization is commonly used as a feature selection criterion. It penalizes irrelevant parameters by shrinking their weights or coefficients to zero. Hence, those features are removed from the model, and it not only removes the extraneous features and prevents the model from overfitting.

Random Forest Feature Importance

Random Forest falls under ensemble learning algorithms that utilize several weak learners' aggregation (decision trees) for prediction. This tree-based approach naturally ranks the features of a dataset by measuring how well the purity is improving. In the decision trees, the impurity drops rapidly at the starting node of the tree, and this rate decreases as we go down. Naturally, the initial node of the tree holds more critical information. Hence, such features are relevant from the feature selection perspective, while those contributing to the lower portion of the tree are less relevant.

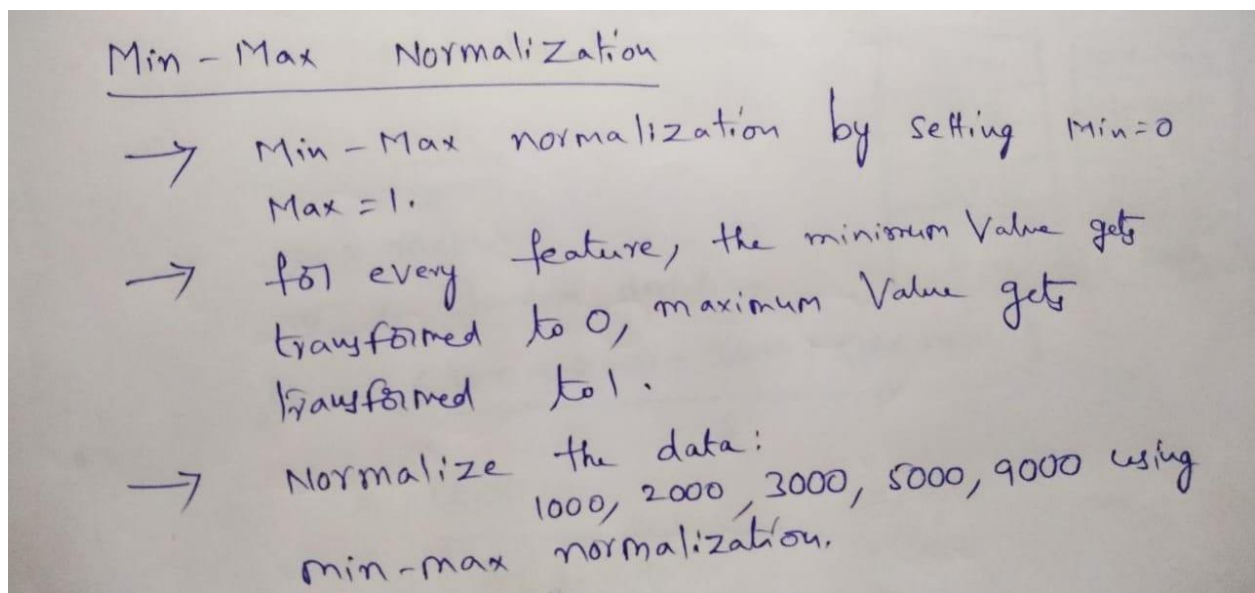
Feature Normalization techniques

- Feature normalization is to make different features in the same scale.

1.min-max normalization

2.z-score normalization

3.constant factor normalization



Sol: —

Data
1000
2000
3000
5000
9000

$$\text{min} = 1000$$

$$\text{max} = 9000$$

$$V = \frac{x - \text{min}}{\text{max} - \text{min}}$$

$$V = \frac{1000 - 1000}{9000 - 1000} = 0$$

$$V = \frac{2000 - 1000}{9000 - 1000} = 0.125$$

$$V = \frac{3000 - 1000}{9000 - 1000} = 0.25$$

$$V = \frac{5000 - 1000}{9000 - 1000} = 0.5$$

$$V = \frac{9000 - 1000}{9000 - 1000} = 1$$

Normalized

0
0.125
0.25
0.5
1

Constant factor Normalization

$$x' = \frac{x}{K}$$

x is raw Value

x' is normalized Value

K is numeric Constant

Data

28
46
34

if $K=100$,

$\frac{28}{100} = 0.28$
$\frac{46}{100} = 0.46$
$\frac{34}{100} = 0.34$

Normalized data

0.28
0.46
0.34

Z-Score Normalization

$$Z = \frac{x - \mu}{\sigma}$$

μ = mean

σ = standard deviation

$$\mu = \left(\frac{1000 + 2000 + 3000 + 5000 + 9000}{5} \right)$$
$$= 4000$$

Data	Normalized
1000	-1.204
2000	-0.803
3000	-0.4016
5000	0.4016
9000	2.008

$$\sigma = \sqrt{\frac{\sum (x_i - \mu)^2}{n-1}}$$
$$= \sqrt{\frac{(1000 - 4000)^2 + (2000 - 4000)^2 + (3000 - 4000)^2 + (5000 - 4000)^2 + (9000 - 4000)^2}{5-1}}$$
$$= 2489.97$$

$$Z = \frac{x - \mu}{\sigma}$$

$$V = \frac{1000 - 4000}{2489.97} = -1.204$$

$$V = \frac{2000 - 4000}{2489.97} = -0.803$$

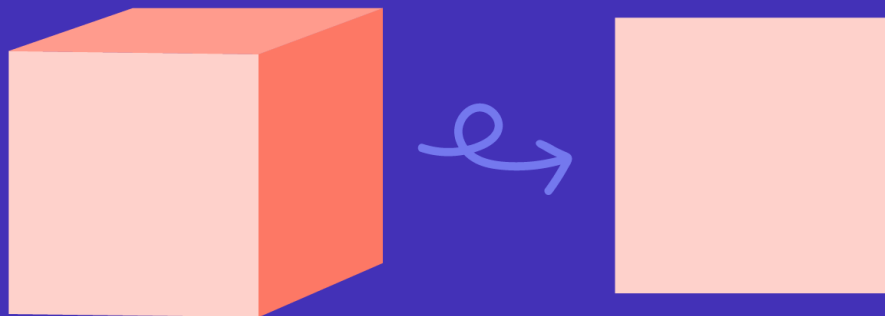
$$V = \frac{3000 - 4000}{2489.97} = -0.4016$$

$$V = \frac{5000 - 4000}{2489.97} = 0.4016$$

$$V = \frac{9000 - 4000}{2489.97} = 2.008$$

Dimensionality Reduction

- Dimensionality reduction is the process of reducing the number of features (or dimensions) in a dataset while retaining as much information as possible.
- This can be done for a variety of reasons, such as to reduce the complexity of a model, to improve the performance of a learning algorithm, or to make it easier to visualize the data.
- There are several techniques for dimensionality reduction, including
 - 1.principal component analysis (PCA)
 - 2.linear discriminant analysis (LDA)
- Transpose of original data and multiply it by transposing of the derived feature vector. Reducing the features without losing information.



1.principal component analysis (PCA)

The steps involved for PCA are as follows-

1. Original Data
2. Normalize the original data (mean =0, variance =1)
3. Calculating covariance matrix
4. Calculating Eigen values, Eigen vectors, and normalized Eigenvectors
5. Calculating Principal Component (PC)
6. Plot the graph for orthogonality between PCs

consider dataset,

Feature	sample1	sample2	sample3	sample4
a	4	8	13	7
b	11	4	5	14

Step 1:

No. of features, $D = 2$ (a, b)

No. of samples, $N = 4$ (sample1, sample2, sample3, sample4)

Step 2:

calculating mean,

$$\bar{a} = \frac{4+8+13+7}{4} = 8$$

$$\bar{b} = \frac{11+4+5+14}{4} = 8.5$$

Step 3:

calculating covariance matrix, between features,

In the given dataset, ordered features are as,

(a, a), (a, b), (b, a), (b, b)

$$\begin{aligned}\text{cov}(a, a) &= \frac{1}{N-1} \sum_{k=1}^N (a_i - \bar{a})(a_i - \bar{a}) \\&= \frac{1}{N-1} \sum_{k=1}^N (a_i - \bar{a})^2 \rightarrow \text{for same feature} \\&= \frac{1}{4-1} [(4-8)^2 + (8-8)^2 + (13-8)^2 + (7-8)^2] \\&= \frac{4^2 + 0 + 5^2 + 1^2}{3} = \frac{16 + 0 + 25 + 1}{3} \\&= \frac{42}{3} = \underline{\underline{14}}.\end{aligned}$$

$$\begin{aligned}\text{cov}(a, b) &= \frac{1}{N-1} \sum_{k=1}^N (a_i - \bar{a})(b_i - \bar{b}) \\&= \frac{1}{4-1} [(4-8)(11-8.5) + (8-8)(4-8.5) + \\&\quad (13-8)(5-8.5) + (7-8)(14-8.5)] \\&= \frac{1}{3} [(-4)(2.5) + (0) + (5)(-3.5) + (-1)(5.5)] \\&= \frac{1}{3} [-10 - 17.5 - 5.5] = \frac{-33}{3} = \underline{\underline{-11}}\end{aligned}$$

We are calculating means and then calculating the covariance matrix between features.

$$\text{cov}(b, a) = \frac{1}{N-1} \sum_{k=1}^N (b_i - \bar{b})(a_i - \bar{a})$$

$$= \text{cov}(a, b)$$

$$= -11$$

$$\text{cov}(b, b) = \frac{1}{N-1} \sum_{k=1}^N (b_i - \bar{b})(b_i - \bar{b})$$

$$= \frac{1}{N-1} \sum_{k=1}^N (b_i - \bar{b})^2$$

$$= \frac{1}{4-1} [(11-8.5)^2 + (4-8.5)^2 + (5-8.5)^2 + (14-8.5)^2]$$

$$= \frac{1}{3} [(2.5)^2 + (-4.5)^2 + (-3.5)^2 + (5.5)^2]$$

$$= \frac{1}{3} [6.25 + 20.25 + 12.25 + 30.25]$$

$$= \frac{69}{3} = \underline{\underline{23}}$$

Hence covariance matrix can be

$$S = \begin{bmatrix} \text{cov}(a, a) & \text{cov}(a, b) \\ \text{cov}(b, a) & \text{cov}(b, b) \end{bmatrix} = \begin{bmatrix} 14 & -11 \\ -11 & 23 \end{bmatrix}$$

Step 4:-

calculate Eigen value, Eigen vectors, Normalized Eigen vectors.

In order calculate Eigen value,

$$\det(S - \lambda I) = 0$$

$$I (\text{Identity matrix}) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\lambda I = \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

$$\det \left(\begin{bmatrix} 14 & -11 \\ -11 & 23 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right) = 0$$

$$\det \begin{pmatrix} 14-\lambda & -11 \\ -11 & 23-\lambda \end{pmatrix} = 0.$$

$$(14-\lambda)(23-\lambda) - (-11 \times -11) = 0$$

$$322 - 14\lambda - 23\lambda + \lambda^2 - 121 = 0$$

After finding covariance matrix, we are going to calculate the eigenvalue, eigenvector, and normalized eigenvector

$$201 - 37\lambda + \lambda^2 = 0.$$

After rearranging,

$$\lambda^2 - 37\lambda + 201 = 0.$$

' λ ' can be calculated by quadratic eqn,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$a = 1$
 $b = -37$
 $c = 201$

$$= \frac{-(-37) \pm \sqrt{(-37)^2 - 4(1)(201)}}{2(1)}$$

$$= \frac{37 \pm \sqrt{1369 - 804}}{2} = \frac{37 \pm \sqrt{565}}{2}$$

$$= \frac{37 \pm 23.76}{2} \Rightarrow \frac{37 + 23.76}{2}, \frac{37 - 23.76}{2}$$

$$= \frac{60.76}{2}, \frac{13.24}{2}$$

Eigen Values. $\boxed{\lambda_1 = 30.38, \lambda_2 = 6.62}$

So, while arranging in descending order,

$$\lambda_1 > \lambda_2 > \dots$$

Hence, $\lambda_1 = 30.38$
 $\lambda_2 = 6.62$

We are going to find out Eigenvectors for Eigen Value, $\lambda = 30.38$.

$$(S - \lambda_1 I) \vec{v}_1 = 0$$

Covariance matrix
30.38
Identity matrix

Eigenvectors at λ_1

$$\left(\begin{pmatrix} 14 & -11 \\ -11 & 23 \end{pmatrix} - 30.38 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right) \vec{v}_1 = 0$$

Assume $\vec{v}_1 = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$

Hence,

$$\begin{pmatrix} 14 & -11 \\ -11 & 23 \end{pmatrix} - \begin{pmatrix} 30.38 & 0 \\ 0 & 30.38 \end{pmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Steps involved in eigenvalues and vectors, in the manual approach

$$\begin{pmatrix} 14 - 30.38 & -11 \\ -11 & 23 - 30.38 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{pmatrix} -16.38 & -11 \\ -11 & -7.38 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$-16.38 u_1 - 11 u_2 = 0 \rightarrow \textcircled{1}$$

$$-11 u_1 - 7.38 u_2 = 0 \rightarrow \textcircled{2}$$

So, from this if we need to calculate u_1 & u_2

$$\textcircled{1} \times 7.38 \Rightarrow 120.88 u_1 - 81.18 u_2 = 0$$

$$\textcircled{2} \times -11 \Rightarrow +121 u_1 + 81.18 u_2 = 0$$

$$0.12 u_1 = 0$$

$$\boxed{u_1 = 0}$$

then, apply u_1 in $\textcircled{1}$, then

$$-16.38 \times 0 - 11 u_2 = 0$$

$$\boxed{u_2 = 0}$$

this can't be possible, hence

$$\begin{bmatrix} (14 - \lambda_1) & (-11) \\ (-11) & (23 - \lambda_1) \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$(14 - \lambda_1) u_1 - 11 u_2 = 0 \rightarrow (a)$$

$$-11 u_1 + (23 - \lambda_1) u_2 = 0 \rightarrow (b)$$

from (a),

$$(14 - \lambda_1) u_1 - 11 u_2 = 0$$

$$(14 - \lambda_1) u_1 = 11 u_2$$

$$\frac{u_1}{11} = \frac{u_2}{14 - \lambda_1} = A \text{ (Assigning)}$$

Assume $A=1$,

$$\frac{u_1}{11} = \frac{u_2}{14 - \lambda_1} = A = 1$$

$$\text{Hence, } \frac{u_1}{11} = 1 \Rightarrow u_1 = 11$$

$$\begin{aligned} \frac{u_2}{14 - \lambda_1} &= 1 \Rightarrow u_2 = 14 - \lambda_1 \\ &= 14 - 30.38 \\ &= -16.38 \end{aligned}$$

Hence Eigenvectors
for $\lambda_1 \Rightarrow$

$$\begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 11 \\ -16.38 \end{bmatrix}$$

From this, we are going to calculate PCs

Then if we want to normalize the eigen vectors,

$$n_1 = \begin{bmatrix} 11 / \sqrt{11^2 + 16.38^2} \\ -16.38 / \sqrt{11^2 + 16.38^2} \end{bmatrix} \quad / \text{dividing by the length.}$$

$$= \begin{bmatrix} \frac{11}{19.73} \\ -16.38/19.73 \end{bmatrix} = \begin{bmatrix} 0.5575 \\ -0.8302 \end{bmatrix}$$

Now, calculate eigen vectors for $\lambda_2 = 6.62$

$$(S - \lambda_2 I) v_2 = 0$$

$$\begin{bmatrix} (14 - \lambda_2) & -11 \\ -11 & (23 - \lambda_2) \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$(14 - \lambda_2) v_1 - 11 v_2 = 0 \rightarrow (c)$$

$$-11 v_1 - (23 - \lambda_2) v_2 = 0 \rightarrow (d)$$

from (c),

$$(14 - \lambda_2) v_1 - 11 v_2 = 0$$

$$(14 - \lambda_2) v_1 = 11 v_2$$

$$\frac{v_1}{11} = \frac{v_2}{14 - \lambda_2} = B \text{ (Assume)}$$

Assume $B = 1$,

$$\frac{v_1}{11} = \frac{v_2}{14 - \lambda_2} = B = 1$$

$$\text{Hence, } \frac{v_1}{11} = 1 \Rightarrow v_1 = 11$$

$$\frac{v_2}{14 - \lambda_2} = 1 \Rightarrow v_2 = 14 - \lambda_2$$

$$= 14 - 6.62$$

$$= 7.38$$

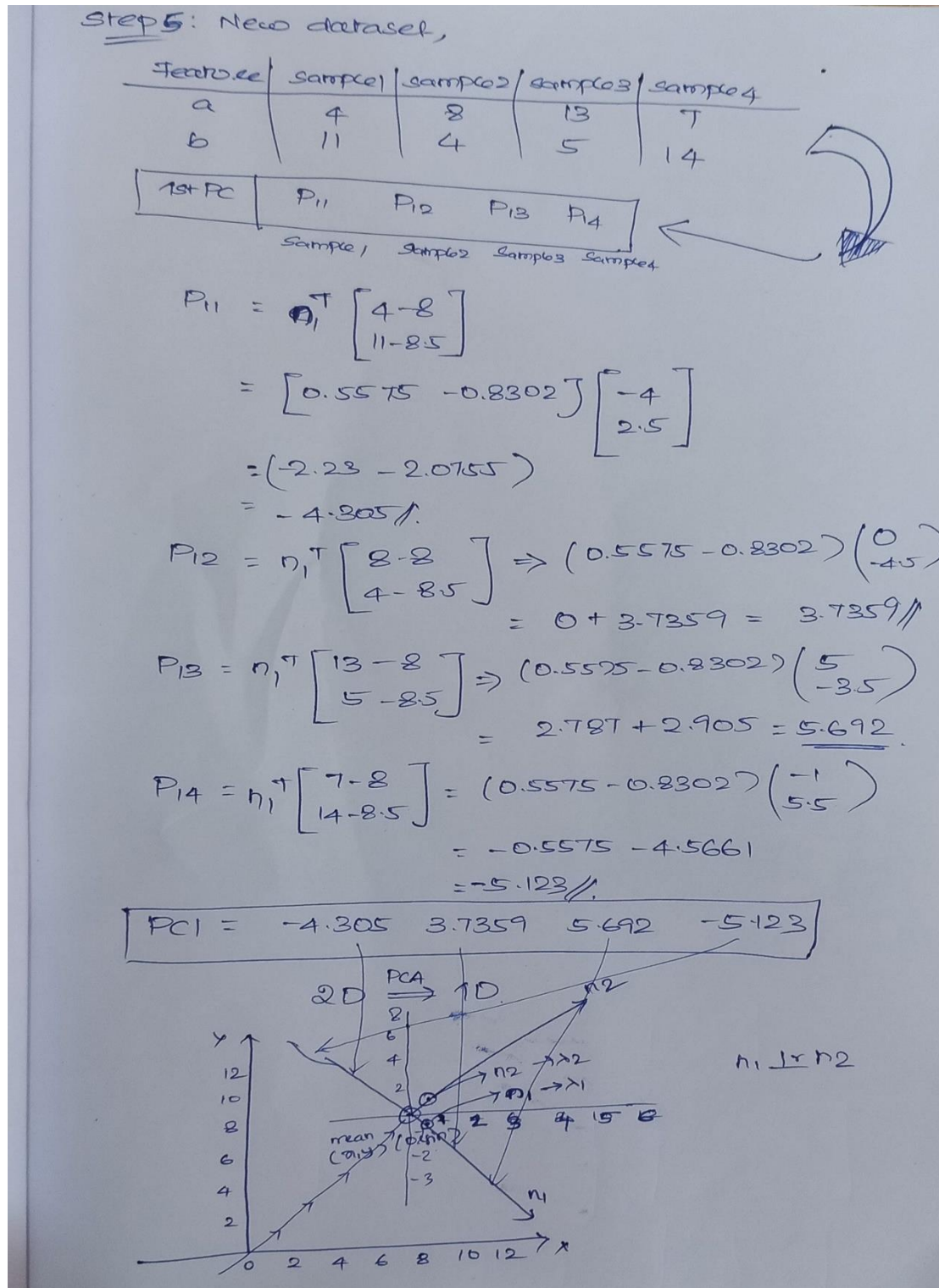
$$\text{Hence, Eigen Vectors for } \lambda_2 = \begin{bmatrix} 11 \\ 7.38 \end{bmatrix}$$

If we want to normalize eigen vectors,

$$n_2 = \begin{bmatrix} 11 / \sqrt{11^2 + 7.38^2} \\ 7.38 / \sqrt{11^2 + 7.38^2} \end{bmatrix} = \begin{bmatrix} 11 / 13.24 \\ 7.38 / 13.24 \end{bmatrix}$$

$$= \begin{bmatrix} 0.8308 \\ 0.5574 \end{bmatrix}$$

We are going to calculate the normalized eigenvector



Hence PCA is calculated and visually we can see how PC are orthogonal to each other.

Advantage for Principal Component Analysis

1. Used for Dimensionality Reduction
2. PCA will assist you in eliminating all related features, sometimes referred to as multi-collinearity.
3. The time required to train your model is now substantially shorter because to PCA's reduction in the number of features.
4. PCA aids in overcoming overfitting by eliminating the extraneous features from your dataset.

Application for Principal Component Analysis

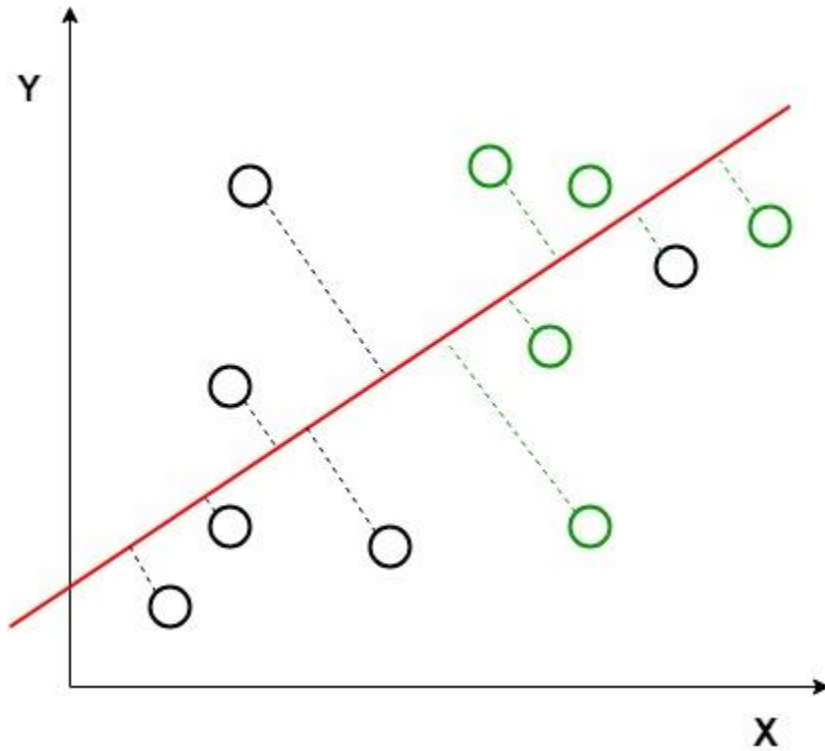
1. Computer Vision
2. Bio-informatics application
3. For compressed images or resizing of the image
4. Discovering patterns from high-dimensional data
5. Reduction of dimensions
6. Multidimensional Data – Visualization

2.Linear Discriminant Analysis (LDA) is a supervised learning algorithm used for classification tasks in machine learning.

It is a technique used to separates the classes in a dataset.

Example:

Suppose we have two sets of data points belonging to two different classes that we want to classify. As shown in the given 2D graph, when the data points are plotted on the 2D plane, there's no straight line that can separate the two classes of the data points completely. Hence, in this case, LDA (Linear Discriminant Analysis) is used which reduces the 2D graph into a 1D graph in order to maximize the separability between the two classes.



In the above graph, it can be seen that a new axis (in red) is generated and plotted in the 2D graph such that it maximizes the distance between the means of the two classes and minimizes the variation within each class. In simple terms, this newly generated axis increases the separation between the data points of the two classes. After generating this new axis using the above-mentioned criteria, all the data points of the classes are plotted on this new axis and are shown in the figure given below.



Linear Discriminant Analysis (LDA) Example Problem

Find the projection dimension that maintain the separation b/w 2 classes.

$$w_1 = \left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix} \begin{pmatrix} 3 \\ 5 \end{pmatrix} \begin{pmatrix} 4 \\ 3 \end{pmatrix} \begin{pmatrix} 5 \\ 6 \end{pmatrix} \begin{pmatrix} 7 \\ 5 \end{pmatrix} \right\}$$

$$w_2 = \left\{ \begin{pmatrix} 6 \\ 2 \end{pmatrix} \begin{pmatrix} 9 \\ 4 \end{pmatrix} \begin{pmatrix} 10 \\ 1 \end{pmatrix} \begin{pmatrix} 12 \\ 3 \end{pmatrix} \begin{pmatrix} 13 \\ 6 \end{pmatrix} \right\}$$

Sol.:- $\mu_1 = \begin{pmatrix} 4 \\ 4.2 \end{pmatrix} \quad \mu_2 = \begin{pmatrix} 10 \\ 3.2 \end{pmatrix}$

Calculate Scatter of class w_1, w_2

$$S_1 = \sum_{x_i \in w_1} (x_i - \mu_1)(x_i - \mu_1)^T$$

$$S_2 = \sum_{x_i \in w_2} (x_i - \mu_2)(x_i - \mu_2)^T$$

$$(x - \mu_1) = \left\{ \begin{matrix} -3 & -1 & 0 & 1 & 3 \\ -2.2 & 0.8 & -1.2 & 1.8 & .8 \end{matrix} \right\}$$

$$S_1 = \begin{pmatrix} -3 & -1 & 0 & 1 & 3 \\ -2.2 & 0.8 & -1.2 & 1.8 & .8 \end{pmatrix} \begin{pmatrix} -3 & -2.2 \\ -1 & 0.8 \\ 0 & -1.2 \\ 1 & 1.8 \\ 3 & .8 \end{pmatrix}$$

$$= \begin{pmatrix} 20 & 10 \\ 10 & 10.8 \end{pmatrix}$$

$$(x - \mu_2) = \begin{pmatrix} -4 & -1 & 0 & 2 & 3 \\ -1.2 & 0.8 & -2.2 & -0.2 & 2.8 \end{pmatrix}$$

$$S_2 = \begin{pmatrix} -4 & -1 & 0 & 2 & 3 \\ -1.2 & 0.8 & -2.2 & -0.2 & 2.8 \end{pmatrix} \begin{pmatrix} -4 & -1.2 \\ -1 & 0.8 \\ 0 & -2.2 \\ 2 & -0.2 \\ 3 & 2.8 \end{pmatrix}$$

$$S_2 = \begin{pmatrix} 30 & 12 \\ 12 & 14.8 \end{pmatrix}$$

Covariance of 2 class on S_1, S_2 .

$$S_W = S_1 + S_2$$

$$S_W = \begin{pmatrix} 50 & 22 \\ 22 & 25.6 \end{pmatrix}$$

$$S_W^{-1} = \frac{\text{adj}(S_W)}{|S_W|}$$

$$= \frac{1}{796} \begin{pmatrix} 25.6 & -22 \\ -22 & 50 \end{pmatrix}$$

$$S_W^{-1} = \begin{pmatrix} .032 & -0.03 \\ -0.03 & .06 \end{pmatrix}$$

$$\begin{aligned} W &= S_W^{-1}(\mu_1 - \mu_2) \\ &= \begin{pmatrix} .032 & -0.03 \\ -0.03 & .06 \end{pmatrix} \left[\begin{pmatrix} 4 \\ 4.2 \end{pmatrix} - \begin{pmatrix} 10 \\ 3.2 \end{pmatrix} \right] \\ &= \begin{pmatrix} -2.2 \\ 2.4 \end{pmatrix} \end{aligned}$$

